

6.4 k -Flows for small k

Trivially, a graph has a 1-flow (the empty set) if and only if it has no edges. In this section we collect a few simple examples of sufficient conditions under which a graph has a 2-, 3- or 4-flow. More examples can be found in the exercises.

Proposition 6.4.1. *A graph has a 2-flow if and only if all its degrees are even.*

Proof. By Theorem 6.3.3, a graph $G = (V, E)$ has a 2-flow if and only if it has a $\mathbb{Z}_2$ -flow, i.e. if and only if the constant map $\vec{E} \rightarrow \mathbb{Z}_2$ with value $\bar{1}$ satisfies (F2). This is the case if and only if all degrees are even. $\square$ (6.3.3)

For the remainder of this chapter, let us call a graph *even* if all its vertex degrees are even. *even graph*

Proposition 6.4.2. *A cubic graph has a 3-flow if and only if it is bipartite.*

Proof. Let $G = (V, E)$ be a cubic graph. Let us assume first that G has a 3-flow, and hence also a $\mathbb{Z}_3$ -flow f . We show that any cycle $C = x_0 \dots x_\ell x_0$ in G has even length (cf. Proposition 1.6.1). Consider two consecutive edges on C , say $e_{i-1} := x_{i-1}x_i$ and $e_i := x_i x_{i+1}$. If f assigned the same value to these edges in the direction of the forward orientation of C , i.e. if $f(e_{i-1}, x_{i-1}, x_i) = f(e_i, x_i, x_{i+1})$, then f could not satisfy (F2) at x_i for any non-zero value of the third edge at x_i . Therefore f assigns the values $\bar{1}$ and $\bar{2}$ to the edges of C alternately, and in particular C has even length. (1.6.1) (6.3.3)

Conversely, let G be bipartite, with vertex bipartition $\{X, Y\}$. Since G is cubic, the map $\vec{E} \rightarrow \mathbb{Z}_3$ defined by $f(e, x, y) := \bar{1}$ and $f(e, y, x) := \bar{2}$ for all edges $e = xy$ with $x \in X$ and $y \in Y$ is a $\mathbb{Z}_3$ -flow on G . By Theorem 6.3.3, then, G has a 3-flow. $\square$

What are the flow numbers of the complete graphs K^n ? For odd $n > 1$, we have $\varphi(K^n) = 2$ by Proposition 6.4.1. Moreover, $\varphi(K^2) = \infty$, and $\varphi(K^4) = 4$; this is easy to see directly (and it follows from Propositions 6.4.2 and 6.4.5). Interestingly, K^4 is the only complete graph with flow number 4:

Proposition 6.4.3. *For all even $n > 4$, $\varphi(K^n) = 3$.*

Proof. Proposition 6.4.1 implies that $\varphi(K^n) \geq 3$ for even n . We show, by induction on n , that every $G = K^n$ with even $n > 4$ has a 3-flow. (6.3.3)

For the induction start, let $n = 6$. Then G is the edge-disjoint union of three graphs G_1, G_2, G_3 , with $G_1, G_2 = K^3$ and $G_3 = K_{3,3}$. Clearly G_1 and G_2 each have a 2-flow, while G_3 has a 3-flow by Proposition 6.4.2. The union of all these flows is a 3-flow on G .

Now let $n > 6$, and assume the assertion holds for $n - 2$. Clearly, G is the edge-disjoint union of a K^{n-2} and a graph $G' = (V', E')$ with $G' = \overline{K^{n-2}} * K^2$. The K^{n-2} has a 3-flow by induction. By Theorem 6.3.3, it thus suffices to find a $\mathbb{Z}_3$ -flow on G' . For every vertex z of the $\overline{K^{n-2}} \subseteq G'$, let f_z be a $\mathbb{Z}_3$ -flow on the triangle $xyz \subseteq G'$, where $e = xy$ is the edge of the K^2 in G' . Let $f: \overline{E'} \rightarrow \mathbb{Z}_3$ be the sum of these flows. Clearly, f is nowhere zero, except possibly in (e, x, y) and (e, y, x) . If $f(e, x, y) \neq \bar{0}$, then f is the desired $\mathbb{Z}_3$ -flow on G' . If $f(e, x, y) = \bar{0}$, then $f + f_z$ (for any z) is a $\mathbb{Z}_3$ -flow on G' . $\square$

Proposition 6.4.4. *Every 4-edge-connected graph has a 4-flow.*

(2.4.2) *Proof.* Let $G = (V, E)$ be a 4-edge-connected graph. By Corollary 2.4.2, G has two edge-disjoint spanning trees $T_i, i = 1, 2$. For each edge $e \notin T_i$ let $C_{i,e}$ be the fundamental cycle with respect to T_i containing e , and let $f_{i,e}$ be a $\mathbb{Z}_4$ -flow of value $\bar{i}$ around $C_{i,e}$ – more precisely: a circulation $\bar{E} \rightarrow \mathbb{Z}_4$ with values $\bar{i}$ and $-\bar{i}$ on the edges of $C_{i,e}$ and zero elsewhere.

f_1 Let $f_1 := \sum_{e \notin T_1} f_{1,e}$. Since each $e \notin T_1$ lies on only one cycle $C_{1,e'}$ (namely, for $e = e'$), f_1 takes only the values $\bar{1}$ and $-\bar{1} (= \bar{3})$ outside T_1 . Let

$$F := \{e \in E(T_1) \mid f_1(e) = \bar{0}\}$$

f_2 and $f_2 := \sum_{e \in F} f_{2,e}$. As above, $f_2(e) = \bar{2} = -\bar{2}$ for all $e \in F$. Now
 f $f := f_1 + f_2$ is the sum of circulations with values in $\mathbb{Z}_4$, and hence itself a circulation with values in $\mathbb{Z}_4$. Moreover, f is nowhere zero: on edges in F it takes the value $\bar{2}$, on edges of $T_1 - F$ it agrees with f_1 (and is hence non-zero by the choice of F), and on all edges outside T_1 it takes one of the values $\bar{1}$ or $\bar{3}$. Hence, f is a $\mathbb{Z}_4$ -flow on G , and the assertion follows by Theorem 6.3.3. $\square$

$\mathbb{Z}_m \times \mathbb{Z}_n$ Our next proposition describes the graphs with a 4-flow in terms of those with a 2-flow. Given integers $m, n \geq 2$, write $\mathbb{Z}_m \times \mathbb{Z}_n$ for the group whose elements are the pairs (a, b) with $a \in \mathbb{Z}_m$ and $b \in \mathbb{Z}_n$ and where $(a, b) + (a', b') := (a + a', b + b')$.

Proposition 6.4.5.

- (i) *A graph has a 4-flow if and only if it is the union of two even subgraphs.*
- (ii) *A cubic graph has a 4-flow if and only if it is 3-edge-colourable.*

(6.3.2) *Proof.* By Corollary 6.3.2 and Theorem 6.3.3, a graph has a 4-flow if and
 (6.3.3) only if it has a $\mathbb{Z}_2^2$ -flow, where $\mathbb{Z}_2^2 := \mathbb{Z}_2 \times \mathbb{Z}_2$. Assertion (i) now follows from Proposition 6.4.1.

(ii) Let $G = (V, E)$ be a cubic graph. We assume first that G has a $\mathbb{Z}_2^2$ -flow f , and define an edge colouring $E \rightarrow \mathbb{Z}_2^2 \setminus \{0\}$. As $a = -a$ for all $a \in \mathbb{Z}_2^2$, we have $f(\vec{e}) = f(\bar{e})$ for every $\vec{e} \in \vec{E}$; let us colour the edge e with this colour $f(\vec{e})$. Now if two edges with a common end v had the same colour, then these two values of f would sum to zero; by (F2), f would then assign zero to the third edge at v . As this contradicts the definition of f , our edge colouring is correct.

Conversely, since the three non-zero elements of $\mathbb{Z}_2^2$ sum to zero, every 3-edge-colouring $c: E \rightarrow \mathbb{Z}_2^2 \setminus \{0\}$ defines a $\mathbb{Z}_2^2$ -flow on G by letting $f(\vec{e}) = f(\bar{e}) = c(e)$ for all $\vec{e} \in \vec{E}$. $\square$

Corollary 6.4.6. *Every cubic 3-edge-colourable graph is bridgeless.* $\square$

6.5 Flow-colouring duality

In this section we shall see a surprising connection between flows and colouring: every k -colouring of a plane multigraph gives rise to a k -flow on its dual, and vice versa. In this way, the investigation of k -flows on arbitrary graphs, not necessarily planar, appears as a natural generalization of the familiar map colouring problems in the plane.

Let $G = (V, E)$ and $G^* = (V^*, E^*)$ be dual plane multigraphs. (This implies that G and G^* are connected; see Chapter 4.6.) For simplicity, let us assume that G and G^* have neither bridges nor loops and are non-trivial. For edge sets $F \subseteq E$, let us write

$$F^* := \{e^* \in E^* \mid e \in F\}. \quad F^*$$

Conversely, if a subset of E^* is given, we shall usually write it immediately in the form F^* , and thus let $F \subseteq E$ be defined implicitly via the bijection $e \mapsto e^*$.

Suppose we are given a circulation g on G^* : how can we employ the duality between G and G^* to derive from g some information about G ? The most general property of all circulations is Proposition 6.1.1, which says that $g(X, \bar{X}) = 0$ for all $X \subseteq V^*$. By Proposition 4.6.1, the bonds $E^*(X, \bar{X})$ in G^* correspond precisely to the cycles in G . Thus if we take the composition f of the maps $e \mapsto e^*$ and g , and sum its values over the edges of a cycle in G , then this sum should again be zero. Our first aim is to formalize and prove this observation.

Of course, there is still a technical hitch: since g takes its arguments not in E^* but in $\vec{E}^*$, we cannot simply define f as above: we first have to refine the bijection $e \mapsto e^*$ into one from $\vec{E}$ to $\vec{E}^*$, i.e. assign to every $\vec{e} \in \vec{E}$ canonically one of the two orientations of e^* . This will be the purpose of our first lemma. After that, we shall show that f does indeed sum to zero along any cycle in G .